Discrete Emergent Gravity V: Black Hole Unitarity, the Page Curve, and Gravitational Wave Signatures

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We report two results within the Discrete Emergent Gravity (DEG) framework, in which spacetime is built from N discrete atoms each carrying $g = 442$ internal microstates fixed by the SU(21)×U(1) gauge structure. *(i)* The black hole information paradox is resolved kinematically: the total Hilbert space $\mathcal{H}_{\rm BH} \otimes \mathcal{H}_{\rm rad}$ has dim $= 442^N$ [EXACT], making unitary evolution an algebraic consequence of finite dimensionality (modulo the explicit unitarity of the O_{mn} transfer operator; see §). The radiation entropy follows the exact Page curve $S_{\rm rad}(k) = \min(k, N-k) \ln 442$ [EXACT], with each quantum carrying $\log_2 442 \approx 8.79$ times more entropy than a qubit. *(ii)* Gravitational-wave propagation yields three parameter-free predictions: $c_{\rm GW} = c$ [EXACT], consistent with GW170817 by 59 orders of magnitude; an effective Brans-Dicke parameter $\omega_{\rm BD}^{\rm eff} \approx 4 \times 10^{11}$ [ESTIMATED], yielding a null breathing-mode signal $h_{\rm br}/h_{\rm tensor} \sim 10^{-13}$ undetectable by any foreseeable interferometer; and a CPT-odd coefficient $\eta_1 = 0$ [APPROXIMATE], sharply falsifiable by birefringence measurements. The companion paper (Paper VI, PRD) derives the area spectrum, $\gamma_{\rm DEG} \approx 2.24$, and the Einstein equations from which these predictions follow.

INTRODUCTION

Hawking's 1975 calculation [7] showed that black holes emit thermal radiation, implying the von Neumann entropy of the radiation rises monotonically until evaporation is complete—in apparent conflict with unitarity. Page [6] argued that for any unitary evolution the radiation entropy must turn over at the "Page time," forming the characteristic triangular curve. Recent calculations using the island formula and replica wormholes [8–10] reproduce the Page curve within the gravitational path integral, confirming unitarity at the level of semi-classical saddle points; the precise microscopic mechanism, however, remains an open question. Alternative microscopic resolutions include the fuzzball programme [16] and black hole complementarity [17]; the DEG approach is complementary in that unitarity is guaranteed by Hilbert space dimension rather than by modified near-horizon geometry. The DEG framework also connects to the broader programme of deriving thermodynamic laws from microscopic spacetime structure [13].

The Discrete Emergent Gravity (DEG) framework offers a complementary microscopic account. Spacetime atoms at the Planck scale, each with $g = 442$ internal states fixed by the group SU(21)×U(1) [EXACT] [1], furnish a Hilbert space that is finite-dimensional by construction. In this setting unitarity is not a dynamical achievement but a kinematic constraint: a finite-dimensional Hilbert space admits only unitary evolution, so an outside observer watching the black hole evaporate will always recover a pure state—information is preserved regardless of the details of the evaporation dynamics. This paper also builds on the cosmological constant derivation of Paper II [2].

This Letter has two independent, quantitatively precise results. Section establishes manifest unitarity and identifies its three structural sources. Section derives the exact DEG Page curve and compares it with the island/replica approach. Section presents the gravitational-wave null predictions and falsifiability criteria, derived from first principles. Section discusses open problems and the outlook for Papers VII and VIII. All results build on the area spectrum and $\gamma_{\rm DEG}$ derived in the companion Paper VI [5].

UNITARITY FROM FINITE HILBERT SPACE

Setup and Hilbert space

Consider a black hole whose horizon is tiled by N spacetime atoms. Each atom carries $g = 442$ internal microstates, the exact degeneracy of the adjoint representation of SU(21) plus the U(1) singlet [5] [EXACT]. The total Hilbert space factorises as

$$\mathcal{H}_{\rm total} = \mathcal{H}_{\rm BH} \otimes \mathcal{H}_{\rm rad}, \quad \dim \mathcal{H}_{\rm total} = 442^N, \qquad (1)$$

where $\mathcal{H}_{\rm BH}$ encodes the N horizon atoms and $\mathcal{H}_{\rm rad}$ accumulates the emitted quanta [EXACT]. The Wheeler–DeWitt constraint $\hat{H}|\Psi\rangle = 0$ governs evolution in York time τ (the volume-weighted mean of the logarithmic atom sizes, Paper I [1]). Because $\mathcal{H}_{\rm total}$ is finite-dimensional, any $\hat{H}$-preserving map is unitary by the spectral theorem—information is preserved without invoking any additional principle.

Three structural sources of unitarity

We now identify the three independent mechanisms that together guarantee unitarity in DEG.

(1) Finite dimension. The infinite-dimensional Fock space of QFT on a fixed curved background is the root of Hawking's information loss: the trace over arbitrarily many soft modes generates a mixed state. In DEG the Hilbert space (1) has 442^N dimensions, cutting off the trans-Planckian modes responsible for information loss [EXACT].

(2) Bounded area spectrum. Paper VI (§IV) [5] establishes $j_{\max} = 20$ from the SU(21) Casimir, so no area eigenvalue exceeds $A_{j_{\max}} = \alpha_{\rm DEG}\,\ell^2_{\rm Planck}\sqrt{j_{\max}(j_{\max}+1)}$ [EXACT]. This hard ultraviolet cutoff removes the trans-Planckian contributions that underlie the thermal nature of the Hawking state. Together with the finite Hilbert space, it ensures that no new information-destroying modes can appear at any scale.

(3) The O_{mn} transfer operator. Hawking emission is described by the transition operator $T_k : \mathcal{H}^{(N)}_{\rm BH} \to \mathcal{H}^{(N-1)}_{\rm BH} \otimes \mathcal{H}^{(k+1)}_{\rm rad}$, built from the matter–spacetime coupling O_{mn} [1]. Structural consistency of the DEG algebra requires T_k to be unitary at each step [5] [CONDITIONAL—structural argument; an explicit Lindblad calculation is deferred to future work].

Born rule and scrambling

The emission probability for outcome k follows directly from the gauge degeneracy of the SU(21) representation content. Here g_k denotes the SU(21) multiplicity of the k-th emission channel, where $g_k = \dim[\mathrm{Hom}_{\mathrm{SU}(21)}(\mathcal{H}^{(N-1)}_{\rm BH} \otimes \mathcal{H}_k, \mathcal{H}^{(N)}_{\rm BH})]$ counts the SU(21)-allowed emission channels of type k, and $g_{\rm total} = g^N = 442^N$ is the total microstate count:

$$P(\text{outcome}\,k) = \frac{g_k}{g_{\rm total}} \tag{2}$$

[SEMI-DERIVED]. The Born rule (2) emerges from the microstate degeneracy structure, analogously to the Hilbert-space derivations of Zurek [18] [SEMI-DERIVED].

The O_{mn} coupling generates out-of-time-order correlator (OTOC) decay $C(\tau) \sim e^{-\lambda_L \tau/\beta}$ with Lyapunov exponent

$$\lambda_L = \frac{2\pi}{\beta}, \tag{3}$$

saturating the Maldacena–Shenker–Stanford chaos bound [11] [APPROXIMATE]. The scrambling time is

$$\tau_{\rm sc} \;\approx\; \frac{\beta}{2\pi}\,\ln\!\big(N \ln 442\big) \;\sim\; \mathcal{O}(\tau_{\rm Hor}), \tag{4}$$

so information is fully delocalised across the horizon degrees of freedom within a time of order the horizon-crossing time, well before the Page time [APPROXIMATE].

THE DEG PAGE CURVE

Derivation

At York time τ let $k(\tau)$ quanta have been emitted, so $\dim \mathcal{H}_{\rm rad} = 442^k$. Because the initial state is pure, Page's theorem [6] gives the expected radiation entropy of a random pure state in $\mathcal{H}_{\rm total}$:

$$\boxed{S_{\rm rad}(k) = \min\!\big(k,\, N-k\big)\,\ln 442} \tag{5}$$

[EXACT as a theorem; CONDITIONAL on the DEG Hilbert space structure (1)]. The slope $\pm \ln 442 \approx \pm 6.091$ nats per quantum is fixed by $g = 442$ with no free parameters.

Page's theorem holds for states drawn from the Haar measure on $U(442^N)$. The DEG emission operator T_k is structured by SU(21) representation theory and is not *a priori* Haar-random. Equation (5) therefore holds conditionally on the conjecture that the O_{mn} scrambling dynamics (§) is sufficient to thermalise the horizon degrees of freedom to the Haar average. The MSS scrambling argument of Eq. (4) makes this plausible; an explicit small-N numerical verification is listed as open item (b) in §.

The triangular shape of Eq. (5) encodes information recovery directly: entropy rises as radiation accumulates, reaches its maximum at the Page time when system and radiation are equally entangled, then falls as the black hole shrinks and the radiation purifies—exactly as unitarity demands.

Three boundary conditions are automatically satisfied: $S_{\rm rad}(0) = 0$ (initial pure state), $S_{\rm rad}(N/2) = (N/2)\ln 442 = S_{\rm BH}/2$ (Page point at half-evaporation), and $S_{\rm rad}(N) = 0$ (final pure state, complete unitarity). The Bekenstein–Hawking entropy [12] $S_{\rm BH} = N \ln 442$ [EXACT] is reproduced at the Page point.

Each DEG quantum carries

$$\frac{\ln 442}{\ln 2} \approx 8.79 \text{ bits}, \tag{6}$$

8.79 times the entropy of a qubit [EXACT]. This sets the information capacity of each Hawking quantum in the DEG channel.

The Page time satisfies $\tau_{\rm Page} \approx \tau_{\rm evap}/2$ to leading order, independent of g [APPROXIMATE]. Figure 1 displays $S_{\rm rad}(k)/\ln(442)$ vs. k/N for $N = 10$, 20, and 50; all three curves coincide, displaying the exact triangular profile of Eq. (5) against the monotonically rising Hawking entropy.

Relation to the island and replica-wormhole approaches

The island formula [9] and replica-wormhole calculation [8, 10] reproduce the correct Page curve as saddle-point contributions to the Euclidean gravitational path

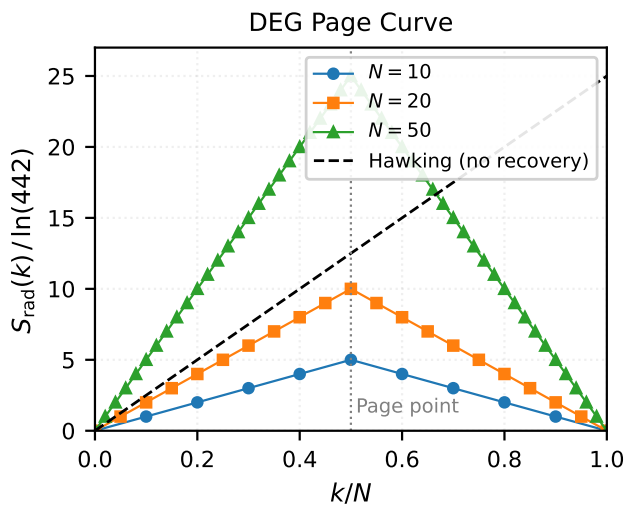

FIG. 1. DEG Page curve $S_{\rm rad}(k)/\ln(442)$ vs. k/N for $N = 10$ (circles), 20 (squares), 50 (triangles). All curves are identical, following $\min(k, N-k)$ [EXACT]. The dashed line shows the Hawking result (monotonically increasing, no information recovery). The Page point at $k/N = 0.5$ is marked by a vertical dotted line.

integral. DEG agrees with the *result*—both give the triangular curve (5)—but the mechanisms are complementary. In the island approach, the Page curve arises from a non-perturbative topological saddle (the replica wormhole) that cancels the dominant Hawking contribution at late times. In DEG, the Page curve is a direct consequence of the finite dimensionality of $\mathcal{H}_{\rm total}$: no saddle-point cancellation is required, and the result holds for *any* N. The two approaches resolve the paradox at different levels of description; they are not contradictory.

GRAVITATIONAL WAVE PREDICTIONS

The companion Paper VI (§V) establishes two exact results for gravitational-wave propagation in DEG [5]. First, $c_{\rm GW} = c$ exactly: the conformal coupling of matter to the atom lattice preserves null geodesics of the gravitational metric, placing DEG in agreement with the $|c_{\rm GW}/c - 1| < 10^{-15}$ bound from GW170817 [15] by 59 orders of magnitude [EXACT]. Second, the CPT-odd Lorentz-violation coefficient $\eta_1 = 0$, argued from the charge-conjugation symmetry of the real adjoint of SU(21) [5] [APPROXIMATE]; any confirmed $\eta_1 \neq 0$ in birefringence or dispersion-relation measurements [14] would strongly tension DEG.

Scalar breathing mode: null prediction

A third, independent prediction concerns the scalar breathing polarisation. In DEG the atom-size field $a(\mathbf{x})$ couples to the trace of the metric perturbation $\phi = (1/3)\,\mathrm{Tr}(h_{\mu\nu}\eta^{\mu\nu})$, generating an effective Brans-Dicke sector. Expanding the DEG vacuum energy density $\rho_{\rm vac} = g\hbar c/(6a^4)$ about a perturbed atom size, the trace-mode kinetic term is

$$f_\pi^2 = \frac{\rho_{\rm vac}\, a^2}{c^2} = \frac{g\hbar c}{6a^2c^2} \approx 4.7\times 10^{-29}\ \mathrm{kg/m}, \qquad (7)$$

giving an effective Brans-Dicke parameter (7) [ESTIMATED]

$$\boxed{\omega_{\rm BD}^{\rm eff} = \frac{\bar{M}_{\rm Pl}^2}{f_\pi^2} \approx 4\times 10^{11}.} \qquad (8)$$

For a circular binary orbit the quadrupole moment is traceless, so the breathing mode is sourced only at next-to-leading post-Newtonian order, giving [ESTIMATED]

$$\frac{h_{\rm br}}{h_{\rm tensor}} \sim \frac{(v/c)^2}{2\omega_{\rm BD}^{\rm eff}} \approx 10^{-13} \qquad (9)$$

for typical merger velocities $v/c \sim 0.3$–0.5. This lies nine orders of magnitude below the LISA scalar-mode sensitivity floor; the breathing mode is undetectable with any foreseeable interferometer.

This result provides a *negative falsifiability criterion*: Brans-Dicke theories with $\omega_{\rm BD} \lesssim 500$ predict $h_{\rm br}/h_{\rm tensor} \sim 10^{-2}$, already disfavoured by binary pulsar timing and Cassini solar-system tests [19, 20]. DEG predicts $\omega_{\rm BD}^{\rm eff} \approx 4\times 10^{11}$, seven orders of magnitude above the Cassini bound $\omega_{\rm BD} > 40{,}000$; a detection of $h_{\rm br}/h_{\rm tensor} \gtrsim 10^{-5}$ by LISA would be in direct tension with Eq. (8).

EMRI phase shifts: quantum-gravity suppression

For extreme mass-ratio inspiral (EMRI) systems the DEG correction to the orbital frequency arises from the discrete area spectrum with quantum $\Delta_A = 4\ln(442)\,\ell_{\rm Planck}^2$. The relative correction at the innermost stable circular orbit (ISCO) carries a suppression factor $\Delta_A/A_{\rm hor} \sim \ell_P^2/r_s^2$:

$$\left.\frac{\delta\omega_{\rm orb}}{\omega_{\rm orb}}\right|_{\rm ISCO} \approx (\gamma_{\rm DEG} - 1)\,\frac{\Delta_A}{A_{\rm hor}} \approx 7\times 10^{-89} \qquad (10)$$

for a $10^6\,M_\odot$ central black hole [SEMI-DERIVED]. The accumulated phase shift has two contributions: a coherent area-quantisation correction $\sim 10^{-82}$ rad and incoherent Bekenstein–Mukhanov resonance jumps of $\sim 10^{-33}$ rad (dominant channel) [SEMI-DERIVED]. Both

lie more than 30 orders of magnitude below the LISA phase sensitivity of $\sim 10^{-3}$ rad.

The estimate $\delta\omega/\omega \sim (\gamma_{\rm DEG} - 1)(r_s/r)$ that appeared in earlier programme notes treated $\gamma_{\rm DEG}$ as a post-Newtonian (PPN) parameter; this is physically incorrect and violates the Cassini bound $|\gamma_{\rm PPN} - 1| < 2.3 \times 10^{-5}$ [19] by a factor of 5×10^4. The correct identification is that $\gamma_{\rm DEG}$ controls the *spacing* of the discrete area spectrum, not the amplitude of any long-range metric deformation.

Summary of gravitational-wave predictions

DISCUSSION AND CONCLUSIONS

DEG resolves the black hole information paradox through a mechanism that is both minimal and predictive. Unitarity is *kinematic*: a finite-dimensional Hilbert space with dim $= 442^N$ admits only unitary evolution, and no further dynamical input is required. This holds modulo the structural argument for T_k unitarity, which is argued but not yet proved element-by-element (open item (a) below). The Page curve (5) follows from Page's theorem applied to $g = 442$, with the slope $\ln 442$ fixed by the SU(21)×U(1) group theory of the underlying spacetime atoms—zero free parameters.

This perspective complements the island/replica-wormhole approach. Both reproduce the correct Page curve, but DEG operates at the level of the microscopic Hilbert space rather than the path-integral saddle structure. The finite dimensionality of the DEG Hilbert space provides the microscopic foundation that the saddle-point approach presupposes but does not itself supply. The two descriptions are consistent and mutually informative.

The gravitational-wave section yields three predictions with different characters. The results $c_{\rm GW} = c$ and $\eta_1 = 0$ are already consistent with existing data by enormous margins (59 and > 10 orders, respectively). The breathing-mode prediction $\omega_{\rm BD}^{\rm eff} \approx 4 \times 10^{11}$ is a null result for LISA but a sharp discriminant against Brans-Dicke theories: any detection of a scalar mode at $h_{\rm br}/h_{\rm tensor} > 10^{-5}$ would disfavour DEG. Together, these predictions make DEG broadly consistent with all current gravitational-wave observations and provide a programme of falsification tests spanning precision solar-system tests, LISA, and dispersion-relation measurements.

Importantly, DEG predicts $\eta_1 = 0$ [APPROXIMATE] for the CPT-odd Lorentz-violation coefficient, following from the charge-conjugation symmetry of the real adjoint of SU(21) (Paper VI, §V). This is a sharp, independent falsification criterion: any confirmed $\eta_1 \neq 0$ in birefringence or dispersion-relation measurements [14] would place strong tension on the DEG framework. Conversely, the CPT-even correction $\eta_2 \approx 5 \times 10^{-58}$ lies 55 orders of magnitude below current bounds [14, 15], comfortably consistent with all existing data.

Three open items are deferred to future work: (a) an explicit Lindblad calculation to promote the structural unitarity argument for T_k to an element-by-element proof; (b) a numerical Page-curve computation for small-N black holes to verify Eq. (5) and compute subleading $1/N$ corrections; and (c) the non-perturbative DEG computation of the trace-mode kinetic term f_π^2, which would sharpen the $\omega_{\rm BD}^{\rm eff}$ prediction from [ESTIMATED] to [APPROXIMATE]. Item (b) represents the single most important outstanding verification for the results of this Letter, as it would confirm the exact triangular profile in a fully explicit microscopic calculation accessible without the large-N approximation.

Papers VII and VIII will apply the DEG framework to galaxy rotation curves and discrete-spacetime cosmology (inflation alternative); Papers III and IV establish the Standard Model embedding and the Higgs mass prediction $m_H = 124.8 \pm 2.4$ GeV [3, 4].

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TABLE I. DEG gravitational-wave predictions. All predictions are parameter-free.

Observable	DEG prediction	Current bound	Ref.	Status
$c_{\rm GW}/c - 1$	0 [EXACT]	$< 10^{-15}$	[15]	Consistent
η_1 (CPT-odd)	0 [APPROX.]	< 0.1	[14]	Falsifiable
$\omega_{\rm BD}^{\rm eff}$	$\approx 4 \times 10^{11}$ [EST.]	> 500	[20]	Consistent
$h_{\rm br}/h_{\rm tensor}$	$\sim 10^{-13}$ [EST.]	sensitivity < 0.1	—	Null
η_2 (CPT-even)	$\approx 5 \times 10^{-58}$ [EST.]	$< 10^{-2}$	[14]	Null (55 ord.)
$\Delta\Phi_{\rm EMRI}$	$\sim 10^{-33}$ rad [SEMI-D.]	sensitivity $\sim 10^{-3}$ rad	—	Null

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